

Notes: ‘n’ represents the dimension of the corresponding problem. ‘NI’ represents the total number of iterations. ‘NF’ represents the number of the function evaluation. ‘T’ represents the running time of the algorithm. ‘ $\|g^*\|$ ’ represents the gradient value for the solution. The values of ‘NI’, ‘NF’, ‘T’ and ‘ $\|g^*\|$ ’ are set as ‘NaN’ if and only if the number of iterations is greater than 2000.

Table 1: Numerical results for Test 1				
Problem/n	SHCG-V1	SHCG-V2	SHCG-V3	SHCG-V4
	NI/NF/T/ $\ g^*\ $	NI/NF/T/ $\ g^*\ $	NI/NF/T/ $\ g^*\ $	NI/NF/T/ $\ g^*\ $
dixmaana/3000	19/83/0.124/1.96e-07	24/88/0.071/4.84e-07	21/81/0.064/9.15e-07	22/83/0.061/6.63e-07
dixmaana/30000	17/86/0.472/3.56e-07	18/87/0.473/4.67e-07	20/86/0.467/9.75e-07	25/94/0.521/7.64e-07
dixmaana/600000	22/97/10.664/9.49e-07	25/98/10.704/4.44e-07	14/84/9.155/6.03e-07	20/96/10.577/4.88e-07
dixmaanb/6000	21/88/0.144/2.93e-07	18/85/0.111/1.28e-07	18/84/0.106/6.07e-07	16/80/0.101/7.42e-07
dixmaanb/15000	26/93/0.278/5.66e-07	24/91/0.272/6.90e-07	20/87/0.258/8.58e-08	23/91/0.279/9.48e-08
dixmaanb/60000	19/89/0.902/9.11e-07	22/92/0.929/1.22e-07	23/93/0.934/6.05e-07	17/83/0.852/7.77e-07
dixmaanc/3000	21/80/0.076/8.82e-07	17/82/0.054/1.09e-07	18/81/0.051/9.54e-07	14/79/0.049/8.38e-07
dixmaanc/150000	23/97/2.660/6.40e-07	20/93/2.551/7.69e-07	21/92/2.519/9.06e-07	24/97/2.684/4.42e-07
dixmaand/300000	15/86/4.689/3.48e-07	19/94/5.162/3.16e-07	19/91/4.921/6.66e-07	29/104/5.656/8.72e-07
dixmaane/1500	297/413/0.160/7.81e-07	363/426/0.147/9.23e-07	315/399/0.135/6.24e-07	357/419/0.143/7.91e-07
dixmaane/3000	431/601/0.384/9.98e-07	536/602/0.384/4.04e-07	478/583/0.370/8.63e-07	477/541/0.347/9.94e-07
dixmaane/15000	820/1054/3.196/8.37e-07	1115/1188/3.609/9.82e-07	825/1039/3.126/7.91e-07	997/1064/3.241/9.81e-07
dixmaanf/1500	270/391/0.151/9.23e-07	312/388/0.127/9.22e-07	265/358/0.119/8.35e-07	363/427/0.153/9.80e-07
dixmaanf/3000	290/409/0.260/7.05e-07	390/459/0.292/9.49e-07	359/466/0.297/7.27e-07	364/428/0.285/9.27e-07
dixmaanf/30000	838/1147/6.317/8.98e-07	1114/1200/6.615/9.18e-07	783/996/5.477/6.94e-07	1627/1723/9.507/7.51e-07
dixmaang/300	128/203/0.035/8.32e-07	202/274/0.025/8.99e-07	158/233/0.022/8.75e-07	279/338/0.032/9.76e-07
dixmaang/1500	250/357/0.133/9.69e-07	341/422/0.137/9.63e-07	332/432/0.141/8.96e-07	502/571/0.183/9.00e-07
dixmaang/15000	1609/2151/6.552/4.16e-07	1825/2118/6.398/9.81e-07	1648/1986/5.959/8.60e-07	NaN/NaN/NaN/NaN
dixmaanh/15000	798/1089/3.376/6.72e-07	1130/1233/3.768/9.94e-07	719/889/2.696/9.80e-07	932/1017/3.073/9.02e-07
dixmaanh/21000	613/885/3.555/7.40e-07	902/991/3.990/9.59e-07	1984/4380/17.426/8.67e-07	1130/1234/4.989/9.71e-07
dixmaanh/30000	874/1178/6.484/8.42e-07	1224/1335/7.368/9.88e-07	1095/1358/7.450/8.46e-07	1157/1249/6.886/9.56e-07
dixmaani/30	217/319/0.025/8.31e-07	216/296/0.007/7.96e-07	203/266/0.007/9.81e-07	257/324/0.009/9.08e-07
dixmaani/90	595/797/0.035/9.48e-07	744/842/0.037/9.04e-07	632/746/0.030/8.92e-07	812/888/0.034/9.46e-07
dixmaani/210	1516/2041/0.141/7.77e-07	1341/1433/0.087/9.06e-07	1242/1452/0.089/8.82e-07	1224/1284/0.090/9.70e-07
dixmaanj/6000	858/1161/1.544/7.17e-07	NaN/NaN/NaN/NaN	745/925/1.147/6.73e-07	NaN/NaN/NaN/NaN
dixmaanl/1500	NaN/NaN/NaN/NaN	NaN/NaN/NaN/NaN	NaN/NaN/NaN/NaN	NaN/NaN/NaN/NaN
dixmaanl/300	1113/1523/0.163/8.07e-07	1812/1983/0.176/9.85e-07	1305/1572/0.125/7.68e-07	NaN/NaN/NaN/NaN
dixmaanl/21000	910/1234/5.003/6.05e-07	1879/1980/8.048/9.63e-07	952/1181/4.829/9.08e-07	NaN/NaN/NaN/NaN
dixmaanl/30000	831/1157/6.318/9.53e-07	1933/2020/11.205/9.80e-07	1141/1394/7.732/9.84e-07	1702/1775/9.882/9.78e-07
dixon3dq/50	524/735/0.012/5.65e-07	498/600/0.006/6.78e-07	488/623/0.008/9.40e-07	592/670/0.008/8.22e-07
dixon3dq/100	936/1260/0.014/9.01e-07	1055/1186/0.013/8.49e-07	936/1144/0.012/8.27e-07	779/841/0.009/8.10e-07
dixon3dq/150	1785/2406/0.025/9.68e-07	1725/1877/0.019/9.73e-07	1815/2184/0.021/8.29e-07	1378/1432/0.014/8.41e-07
dqdrtic/10000	95/226/0.026/6.95e-07	118/339/0.026/5.47e-07	77/228/0.016/5.19e-07	111/262/0.021/5.89e-07
dqdrtic/50000	80/206/0.071/7.68e-07	100/297/0.095/7.28e-07	100/294/0.097/7.26e-07	119/293/0.094/9.05e-07
dqdrtic/150000	95/232/0.283/7.57e-07	103/295/0.348/7.38e-07	76/221/0.266/3.47e-07	117/312/0.375/6.64e-07
dqrtic/5000	54/164/0.180/3.14e-07	55/162/0.166/8.83e-07	56/168/0.171/6.27e-07	61/173/0.189/6.38e-07
edensch/500	30/105/0.016/2.61e-07	49/123/0.012/5.82e-07	44/195/0.018/8.63e-07	53/150/0.016/7.67e-07
edensch/20000	67/385/1.502/7.93e-07	62/267/1.053/8.16e-07	78/502/1.973/4.01e-07	66/229/0.911/8.00e-07
edensch/120000	210/1663/36.174/8.37e-07	127/853/18.533/9.84e-07	69/375/8.189/6.99e-07	121/751/16.324/5.46e-07
eg2/100	236/752/0.022/8.64e-07	345/1456/0.021/8.82e-07	303/1422/0.017/7.68e-07	171/560/0.008/5.65e-07
eg2/200	NaN/NaN/NaN/NaN	1399/12379/0.247/7.97e-07	343/1322/0.028/8.24e-07	431/2338/0.051/8.53e-07
eg2/500	NaN/NaN/NaN/NaN	NaN/NaN/NaN/NaN	NaN/NaN/NaN/NaN	NaN/NaN/NaN/NaN
fletchcr/7000	139/1147/0.086/4.75e-07	295/2731/0.155/4.77e-07	260/2188/0.115/7.02e-07	446/4053/0.215/1.11e-07
freuroth/50	218/1174/0.021/9.11e-07	218/869/0.011/6.00e-07	181/550/0.006/8.05e-07	404/1985/0.021/7.64e-07
freuroth/500	NaN/NaN/NaN/NaN	1054/10047/0.136/3.89e-07	1087/10723/0.145/9.94e-07	903/8118/0.110/9.75e-07
freuroth/5000	NaN/NaN/NaN/NaN	NaN/NaN/NaN/NaN	923/9512/0.649/7.43e-07	NaN/NaN/NaN/NaN
genrose/10000	234/392/0.053/5.52e-07	311/495/0.054/9.75e-07	290/462/0.055/9.99e-07	380/519/0.053/9.64e-07
himmelbg/1000	2/13/0.009/7.55e-30	2/13/0.001/7.54e-30	2/13/0.001/7.55e-30	2/13/0.001/7.55e-30
himmelbg/70000	2/15/0.033/8.37e-43	2/15/0.030/8.34e-43	2/15/0.029/8.37e-43	2/15/0.030/8.37e-43
himmelbg/600000	3/16/0.259/5.32e-26	3/16/0.255/1.39e-26	3/16/0.255/1.30e-26	3/16/0.256/1.61e-26
liarwhd/10000	233/680/0.064/3.52e-07	133/488/0.041/9.39e-07	128/456/0.032/3.94e-07	124/455/0.025/8.36e-07
liarwhd/30000	143/537/0.090/7.87e-07	141/607/0.110/1.95e-07	164/562/0.106/5.06e-07	136/577/0.101/9.63e-07
liarwhd/120000	305/882/0.515/3.70e-07	155/683/0.380/6.83e-07	164/651/0.356/1.51e-07	133/594/0.329/5.64e-07
penalty1/100	18/108/0.018/6.16e-07	19/109/0.002/8.92e-07	19/109/0.002/8.13e-07	19/109/0.003/9.29e-07
penalty1/10000	20/94/3.246/2.94e-09	20/92/3.621/2.06e-07	20/92/3.627/4.52e-07	20/92/3.635/1.83e-07
quartc/5000	54/164/0.175/3.14e-07	55/162/0.162/8.83e-07	56/168/0.172/6.27e-07	61/173/0.182/6.38e-07
tridia/100	361/489/0.012/9.50e-07	349/429/0.005/9.49e-07	251/341/0.003/8.82e-07	295/364/0.004/9.16e-07
tridia/300	591/793/0.009/8.11e-07	735/819/0.009/9.61e-07	623/768/0.008/9.58e-07	512/585/0.007/9.22e-07
tridia/1500	1675/2260/0.044/9.30e-07	1763/1855/0.031/9.51e-07	1960/2355/0.040/9.73e-07	1482/1567/0.027/9.96e-07
woods/5000	196/534/0.044/7.03e-07	219/499/0.020/6.96e-07	158/434/0.018/8.84e-07	300/578/0.029/4.16e-07
woods/300000	230/578/1.329/2.66e-07	252/656/1.502/9.23e-07	210/530/1.210/8.38e-07	300/650/1.537/2.03e-07
woods/1000000	179/436/3.248/4.15e-07	391/826/6.351/5.49e-07	218/648/4.677/5.72e-07	376/750/5.895/6.87e-07
bdexp/1000	2/11/0.010/2.98e-133	2/11/0.001/2.96e-133	2/11/0.001/2.99e-133	2/11/0.001/2.99e-133
bdexp/10000	2/17/0.015/2.62e-61	2/17/0.013/2.62e-61	2/17/0.010/2.62e-61	2/17/0.010/2.62e-61
bdexp/100000	2/12/0.081/4.57e-106	2/12/0.079/4.57e-106	2/12/0.078/4.57e-106	2/12/0.078/4.57e-106
exdenschnf/10000	38/114/0.031/5.03e-07	52/130/0.011/9.91e-07	40/113/0.011/8.83e-07	48/124/0.010/4.67e-07
exdenschnb/3000	25/89/0.007/5.69e-07	34/98/0.004/4.71e-07	26/89/0.003/9.86e-07	47/111/0.003/1.10e-07
exdenschnb/5000000	40/115/4.250/7.22e-07	26/108/3.529/1.91e-07	49/125/4.517/3.21e-07	36/111/3.802/4.17e-07
genquartic/50000	18/87/0.045/5.78e-07	28/97/0.037/6.25e-07	25/93/0.035/3.19e-07	29/97/0.044/8.47e-07
genquartic/100000	19/81/0.061/5.37e-07	28/98/0.073/6.85e-07	21/86/0.067/7.90e-07	24/93/0.058/9.72e-07
biggsb1/250	1282/1688/0.026/7.82e-07	1613/1778/0.020/9.16e-07	1518/1838/0.020/9.91e-07	1156/1214/0.014/7.95e-07
biggsb1/500	NaN/NaN/NaN/NaN	NaN/NaN/NaN/NaN	NaN/NaN/NaN/NaN	NaN/NaN/NaN/NaN
fletcbv3/100	873/1724/0.066/9.53e-07	1681/3820/0.125/7.28e-07	986/2159/0.068/7.87e-07	NaN/NaN/NaN/NaN
nonscomp/10000	NaN/NaN/NaN/NaN	351/679/0.052/7.60e-07	NaN/NaN/NaN/NaN	NaN/NaN/NaN/NaN
nonscomp/70000	NaN/NaN/NaN/NaN	NaN/NaN/NaN/NaN	1774/2707/1.464/8.13e-07	NaN/NaN/NaN/NaN
nonscomp/100000	NaN/NaN/NaN/NaN	NaN/NaN/NaN/NaN	1831/2767/2.101/8.60e-07	NaN/NaN/NaN/NaN
power1/100	1137/1545/0.021/7.45e-07	1431/1582/0.016/9.79e-07	1360/1617/0.016/9.75e-07	1489/1589/0.016/8.07e-07
raydan1/300	137/205/0.008/5.48e-07	273/367/0.007/9.66e-07	173/249/0.005/9.08e-07	356/463/0.009/8.59e-07
raydan2/2000	14/71/0.008/7.00e-07	14/67/0.004/5.34e-07	14/71/0.005/7.00e-07	14/71/0.008/7.00e-07
raydan2/70000	21/102/0.327/2.05e-07	35/144/0.458/2.46e-07	12/72/0.234/2.93e-07	12/72/0.237/2.93e-07
diagonal1/250	1066/8565/0.197/8.36e-07	419/2919/0.063/7.89e-07	550/4644/0.102/8.94e-07	474/3601/0.077/9.33e-07
diagonal2/500	156/268/0.010/4.95e-07	217/329/0.012/9.36e-07	158/264/0.010/9.84e-07	253/382/0.015/6.22e-07
diagonal2/5000	463/741/0.229/9.56e-07	626/956/0.293/8.20e-07	535/842/0.264/7.01e-07	601/866/0.269/9.78e-07
diagonal2/10000	764/1191/0.740/9.71e-07	902/1328/0.827/6.71e-07	696/1043/0.647/5.81e-07	837/1195/0.743/9.18e-07
singx/300	307/812/0.093/5.69e-07	154/413/0.044/7.52e-07	161/438/0.044/9.85e-07	101/278/0.032/7.40e-07
singx/1000	225/528/1.020/4.63e-07	166/439/0.834/2.84e-07	243/620/1.167/7.79e-07	156/435/0.817/4.67e-07
singx/3000	188/460/5.700/7.87e-07	98/260/3.237/8.75e-07	347/769/9.526/4.80e-07	148/405/4.951/5.05e-07
lin/30	17/79/0.030/6.58e-07	17/79/0.011/7.20e-07	17/79/0.006/6.96e-07	17/79/0.006/7.02e-07
lin/400	14/72/0.124/7.97e-07	14/72/0.124/7.96e-07	14/72/0.130/7.97e-07	14/72/0.126/7.97e-07
lin/700	18/81/4.467/2.20e-07	18/81/4.471/2.85e-07	18/81/4.454/2.85e-07	18/81/4.419/2.85e-07
osb2/11	606/1011/0.054/7.30e-07	390/634/0.021/7.23e-07	414/708/0.024/8.22e-07	495/751/0.025/8.82e-07
pen1/50	136/510/0.017/7.03e-07	113/436/0.007/4.22e-07	67/315/0.005/8.38e-07	76/386/0.007/7.72e-07
pen1/100	172/555/0.012/7.82e-07	113/465/0.012/3.78e-07	60/329/0.008/8.95e-07	78/483/0.011/4.23e-07
pen1/600	161/1025/1.741/9.99e-07	91/576/0.969/8.56e-07	117/478/0.808/9.98e-07	95/484/0.812/9.95e-07
pen2/10	1245/3208/0.062/7.65e-07	755/2213/0.031/9.50e-07	846/2413/0.033/7.92e-07	519/1687/0.021/9.01e-07
pen2/50	381/1191/0.023/9.45e-07	304/1032/0.019/9.46e-07	287/921/0.018/6.02e-07	351/1235/0.023/8.77e-07
pen2/90	227/755/0.033/8.46e-07	183/506/0.018/6.33e-07	207/684/0.024/7.77e-07	220/739/0.027/3.70e-07
rosex/300	104/296/0.033/9.51e-07	80/305/0.021/3.26e-07	107/292/0.018/9.86e-07	116/393/0.029/7.55e-07
rosex/1000	99/348/1.007/9.04e-08	87/355/1.026/7.78e-07	107/340/0.997/9.72e-07	74/309/0.907/7.73e-07
trid/400	86/148/0.065/7.24e-07	109/170/0.065/9.73e-07	85/145/0.055/6.10e-07	134/194/0.084/9.32e-07